



Industrial Robotics Standards For Safety And Efficiency

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There have been significant changes in industrial safety standards for robotics. Safety managers, plant managers, and others need to keep pace with the latest codes and regulations. According to the US Occupational Safety and Health Administration (OSHA), ‘machine guarding’ that pertains to machines’ general requirements and general industry (29 CFR 1910.212) consistently falls in the top ten most frequently cited OSHA standards violated in any given year. Thanks to the Robotic Industries Association (RIA) R15.06-2013 standard, proper risk assessments are no longer just a best practice; they are mandatory.

RIA 15.06-2013 harmonises international and US standards. The RIA 15.06-2013 was written to be compliant with international standards already in place in Europe. This standard requires better hazard identification related not only to robotic motion, but also to the task being performed. Additionally, it requires validation and verification of the safety systems employed and requires designs that incorporate protective measures for the robot cell and the operator.

Some of the biggest changes in the RIA 15.06 industrial robot standard include safety-rated motion and allowing advanced programmable safety devices to be used. What this means is that the software will now be allowed for ‘safety-rated’ control of various aspects of the robot’s function, limiting the area in which the robot operates and the speed of robot motion.

The basics of machine guarding risk assessment. The first step for facility/safety professionals is to identify and understand all applicable codes and regulations for their facil-

ity and operation. Next, they should examine the prevailing machine guarding choices for those applications to validate their safety system and its components. Risk may even vary between similar operations, depending upon employee exposure and other factors.

Determining risk. When performing a proper risk assessment, point-of-operation guarding is the most involved aspect. It is easy to place perimeter guarding around the entire process. However, in most situations a machine operator needs to interact with the process by loading or unloading materials (such as metals to be welded) and ‘running’ the machine. Many details must be considered when it comes to this area, including the layout or design of the process and the limits of the system.

Once the severity of the potential hazard has been determined, the frequency or duration of exposure and the possibility of eliminating or limiting exposure can help safety managers choose the proper machine guarding device. Also, using the distance formula identified in OSHA guidelines can help in this selection.

Limiting hazard exposure. Light curtains, laser scanners, and other presence-sensing devices are a commonly used and widely accepted method of machine guarding in manufacturing facilities from Tier 1 automotive to small machine shops and fabrication facilities. With presence-sensing, the automated process ceases once the safety device’s infrared beam is tripped. In many instances, these devices provide acceptable safety. However, these are not always the best choice in all applications, especially after a risk assessment is performed. Coupled with safety interlocks (up to PLe per EN ISO 13849-1 when integrated properly), fast-acting automated barrier doors or roll-up curtains

offer an increased level of protection for point-of-operation guarding.

From EN 954-1 to ISO 13849-1 and EN 62061. One of the biggest regulatory paradigm shifts occurred with this move. ISO 13849-1 provides a clearly defined set of rules to follow when designing the safety system as applied to industrial machine control systems. Officially defined as “safety of machinery, safety-related parts of control systems, general principles for design,” this regulatory shift was made necessary by increasingly complex manufacturing processes using robotic and automated technology.

The ISO 13849-1 standard applies common sense and forces facility managers to validate their safety systems, whereas EN 954-1 only required facilities to apply safety devices (controls) properly, specifying non-programmable, out-of-date technology.

EN ISO 13849-1 is ultimately making a much safer manufacturing environment because it accounts for the regulatory gaps in the older standards. For regulations such as RIA 15.06 and EN ISO 13849-1, it is important to keep up with the latest and greatest safety technologies available to match the right product to the right process.

Creating a smaller manufacturing cell. Due to the nature of a properly interlocked automated barrier door, certain aspects of OSHA’s safety distance formula become moot because there is no depth penetration factor. The safeguard can be placed much closer to the hazardous area, which reduces the manufacturing cell’s footprint. This space savings is a huge benefit in most facilities. The smaller safety zone may also help increase productivity and create a better ergonomic situation for the machine operator by limiting required motion. **EFY**

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